

Differential Equations:

An equation containing the derivatives of one or more dependent variables with respect to one or more independent variables.

e.g. $\frac{d^2y}{dx^2} - x \cdot \frac{dy}{dx} + y = e^x$ ← An equation in terms of x and y
where $x \rightarrow$ independent variable
 $y \rightarrow$ dependent variable
and has derivatives of y :
 $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

e.g. $\frac{dx}{dt} - \frac{dy}{dt} = 2x + y$ ← Here, $t \rightarrow$ independent variable
 $x, y \rightarrow$ dependent variable

* So there can also be multiple dependent variables

↳ Two types of DEs:

i) Ordinary Differential Equations (ODEs):

The derivatives are ordinary derivatives

(like $\frac{dy}{dx}$ or $\frac{d^2y}{dx^2}$).

ii) Partial Differential Equations (PDEs):

The derivatives are partial derivatives

(like $\frac{\partial y}{\partial x}$ or $\frac{\partial^2 y}{\partial x^2}$).

* You won't have to deal with equations with both ordinary and partial derivatives.

Each DE (be it ordinary or partial) constitutes of two parts:

1) The order of the DE

2) The degree of the DE

1) Order of a DE:

The highest order of derivatives in a DE

1st order derivative $\rightarrow \frac{dy}{dx}$

2nd order derivative $\rightarrow \frac{d^2y}{dx^2}$

3rd order derivative $\rightarrow \frac{d^3y}{dx^3}$

and so on.

e.g. $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$ $\leftarrow$ highest order derivative is $\frac{d^2y}{dx^2}$
 $\therefore$ Order of DE = 2

e.g. $\frac{d^4y}{dx^4} + \sin y = 0$ $\leftarrow$ highest order derivative is $\frac{d^4y}{dx^4}$
 $\therefore$ Order of DE = 4

2) Degree of a DE:

The highest power of the highest order of derivatives in a DE.

e.g. $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 = 0$ $\leftarrow$ highest order derivative is $\frac{d^2y}{dx^2}$

Power of highest order derivative = 1

$\therefore$ Degree of the DE = 1

e.g. $(\frac{dy}{dx})^2 + 6y = 0$ $\leftarrow$ highest order derivative is $\frac{dy}{dx}$

Power of highest order derivative = 2

$\therefore$ Degree of the DE = 2

Notation:

$\hookrightarrow$ Leibnitz notation: $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots$

$\hookrightarrow$ Prime notation: $y', y'', y''', \dots$

$\hookrightarrow$ Dot notation: $\dot{y}, \ddot{y}, \dddot{y}, \dots$

$\hookrightarrow$ Numbered notation: $y_1, y_2, y_3, \dots$

$\hookrightarrow$ Subscript notation: $u_{xx}, u_{yy}, u_{xy}, \dots$ $\leftarrow$ can also be used for PDEs

Properties of an ODE:

An n -th order ODE is given as follows:

$$a_n(x, y) \cdot (y_n)^{r_n} + a_{n-1}(x, y) \cdot (y_{n-1})^{r_{n-1}} + \dots + a_2(x, y) \cdot (y_2)^{r_2} + a_1(x, y) \cdot (y_1)^{r_1} + a_0(x, y) \cdot (y)^{r_0} = g(x)$$

$\hookrightarrow r_k$ = power of the k -th derivative y_k

$\hookrightarrow a_k(x, y)$ = coefficient function of the k -th derivative y_k

These ODEs can be classified using the following properties:

i) Linearity of the ODE ii) Homogeneity of the ODE.

I) Linearity of the ODE:

An ODE is said to be linear if:

1) y and its derivatives y' , y'' , ..., $y^{(n)}$ all have power = 1

2) the coefficients of the derivatives $a_0, a_1, \dots, a_n$ all depend on x at the most (and on no other variables).

↳ General form:

note the differences between this and the n -th order ODE shown previously

$$a_n(x) \cdot y_n + a_{n-1}(x) \cdot y_{n-1} + \dots + a_2(x) \cdot y_2 + a_1(x) \cdot y_1 + a_0(x) \cdot y = g(x)$$

↳ If an ODE does not satisfy either (1) or (2) then it's said to be non-linear.

(e.g.) $x^3 \cdot \frac{d^3 y}{dx^3} + x \cdot \frac{dy}{dx} - 5y = e^x$ ← Linear

(e.g.) $(y-x) \cdot dx + 4x \cdot dy = 0$

$$\Rightarrow y - x + 4x \cdot \frac{dy}{dx} = 0$$

$$\Rightarrow 4x \cdot \frac{dy}{dx} + y = x$$
 ← Linear

(e.g.) $(1-y) \cdot y' + 2y = e^x$

↑ Coefficient of y' is y ← Non-linear

(e.g.) $\frac{d^2 y}{dx^2} + \sin y = 0$ ← Non-linear

II) Homogeneity of the ODE:

An ODE is said to be homogeneous if $g(x) = 0$.

Otherwise its non-homogeneous.

* $g(x)$ is the part of the function without y or any of its derivatives.

(e.g.) $2y'' + 3y' - 5y = 0$ $\leftarrow$ Homogeneous

(e.g.) $x^3 \cdot y''' + 6y' + 10y = e^x$ $\leftarrow$ Non-homogeneous

Methods of Solving ODEs:

I) Separation of variables

II) Solving Exact DEs

III) Integrating factors

IV) Solving homogeneous linear ODEs with constant coefficients

V) Undetermined Coefficients

* Solutions of ODEs are equations without any derivatives and only in terms of x and y .

Separation of variables:

↳ For: 1st order ODE

↳ ODE is of the form:

$$\frac{dy}{dx} = g(x) \cdot h(y)$$

↳ ODEs of this form can be solved by rearranging them as follows:

$$\frac{dy}{dx} = g(x) \cdot h(y)$$

$$\Rightarrow \frac{1}{h(y)} \cdot dy = g(x) \cdot dx$$

Bring function of x
and dx to one side
and function of y

Introduce integral $\rightarrow$
sign on both sides
and integrate

$$\Rightarrow \int \frac{1}{h(y)} dy = \int g(x) dx$$

and dy to the
other side

(e.g.) Solve $(1+x) dy - y \cdot dx = 0$

Solⁿ: $\Rightarrow (1+x) dy = y \cdot dx$

$$\Rightarrow \frac{1}{y} \cdot dy = \frac{1}{1+x} \cdot dx$$

Bring terms in y and dy
to one side and terms in
 x and dx to the other

$$\Rightarrow \int \frac{1}{y} \cdot dy = \int \frac{1}{1+x} \cdot dx$$

$$\Rightarrow \ln(y) = \ln(1+x) + \ln C = \ln(C \cdot (1+x))$$

$$\Rightarrow y = C \cdot (1+x) \quad \leftarrow \text{here } C \text{ is a constant.}$$

(e.g.) Solve the initial value problem (IVP):

$$\frac{dy}{dx} = -\frac{x}{y} \quad \text{where } y(4) = -3.$$

Soln

$$\Rightarrow \int y \cdot dy = \int -x \, dx$$

$$\Rightarrow \frac{y^2}{2} = -\frac{x^2}{2} + C$$

This means
at $x=4$, we
have $y=-3$

$$\text{At } x=4, y=-3$$

$$\therefore \frac{(-3)^2}{2} = -\frac{(4)^2}{2} + C$$

$$\Rightarrow \frac{9}{2} = -\frac{16}{2} + C$$

$$\Rightarrow C = \frac{25}{2}$$

$$\therefore \frac{y^2}{2} = -\frac{x^2}{2} + \frac{25}{2} \quad \Rightarrow \quad y^2 + x^2 = 25 //$$

(e.g.) Solve the following ODEs using the method of separating variables:

$$\text{I) } (1+y^2)dx + (1+x^2)dy = 0$$

$$\text{II) } \sec^2 x \cdot \tan y \, dx + \sec^2 y \cdot \tan x \, dy = 0$$

$$\text{III) } (x^2 - yx^2) \cdot dy + (y^2 + xy^2) \cdot dx = 0$$

$$\text{IV) } x(y^2 + 1)dx + y(x^2 + 1)dy = 0$$

$$\text{V) } \frac{dy}{dx} = y^3 \cdot x \cdot e^{x^2}, \quad y(0) = 1$$

Soln I) $(1+y^2)dx + (1+x^2)dy = 0$

$$\Rightarrow (1+x^2)dy = -(1+y^2)dx$$

$$\Rightarrow \frac{1}{1+y^2} dy = \frac{-1}{1+x^2} dx$$

$$\Rightarrow \int \frac{1}{1+y^2} dy = - \int \frac{1}{1+x^2} dx$$

$$\Rightarrow \tan^{-1}(y) = -\tan^{-1}(x) + \tan^{-1}(c)$$

$$\Rightarrow \tan^{-1}y + \tan^{-1}x = \tan^{-1}c$$

$$\Rightarrow \tan^{-1}\left(\frac{x+y}{1-xy}\right) = \tan^{-1}c$$

$$\Rightarrow \frac{x+y}{1-xy} = c //$$

II) $\sec^2 x \cdot \tan y dx + \sec^2 y \cdot \tan x dy = 0$

$$\Rightarrow \sec^2 y \cdot \tan x dy = -\sec^2 x \cdot \tan y dx$$

$$\Rightarrow \frac{\sec^2 y}{\tan y} dy = -\frac{\sec^2 x}{\tan x} dx$$

$$\Rightarrow \int \frac{\sec^2 y}{\tan y} dy = - \int \frac{\sec^2 x}{\tan x} dx$$

$$\Rightarrow \ln(\tan y) = -\ln(\tan x) + \ln C$$

$$\Rightarrow \ln(\tan y) + \ln(\tan x) = \ln C$$

$$\Rightarrow \tan x \cdot \tan y = C //$$

$$\text{III) } (x^2 - yx^2) \cdot dy + (y^2 + xy^2) \cdot dx = 0$$

$$\Rightarrow x^2(1-y) dy + y^2(1+x) dx = 0$$

$$\Rightarrow \frac{(1+x)}{x^2} dx = \frac{(y-1)}{y^2} dy$$

$$\Rightarrow \int x^{-2} + x^{-1} dx = \int y^{-1} - y^{-2} dy$$

$$\Rightarrow \frac{x^{-1}}{-1} + \ln x = \ln y - \frac{y^{-1}}{-1} + C$$

$$\Rightarrow \frac{-1}{x} + \ln x = \ln y + \frac{1}{y} + C //$$

$$\text{IV) } x(y^2+1) dx + y(x^2+1) dy = 0$$

$$\Rightarrow y(x^2+1) dy = -x(y^2+1) dx$$

$$\Rightarrow \int \frac{y}{y^2+1} dy = \int \frac{-x}{x^2+1} dx$$

$$\Rightarrow \frac{1}{2} \int \frac{2y}{y^2+1} dy = -\frac{1}{2} \int \frac{2x}{x^2+1} dx$$

$$\Rightarrow \frac{1}{2} \ln(y^2+1) = -\frac{1}{2} \ln(x^2+1) + \frac{1}{2} \ln C$$

P.T.O

$$\Rightarrow \ln(y^2 + 1) + \ln(x^2 + 1) = \ln C$$

$$\Rightarrow \ln([y^2 + 1][x^2 + 1]) = \ln C$$

$$\Rightarrow (y^2 + 1)(x^2 + 1) = C //$$

$$\text{v)} \frac{dy}{dx} = y^3 \cdot x \cdot e^{x^2}, \quad y(0) = 1$$

$$\Rightarrow \frac{1}{y^3} dy = x \cdot e^{x^2} \cdot dx$$

$$\Rightarrow \int \frac{1}{y^3} dy = \frac{1}{2} \int 2x \cdot e^{x^2} dx$$

$$\Rightarrow \frac{y^{-2}}{-2} = \frac{1}{2} [e^{x^2}] + \frac{C}{2}$$

$$\Rightarrow -\frac{1}{y^2} = e^{x^2} + C \quad \leftarrow \text{Given initial condition at } x=0, y=1.$$

$$\Rightarrow 2 = e^{x^2} + \frac{1}{y^2} \quad \leftarrow \therefore \frac{-1}{(1)^2} = e^0 + C \Rightarrow C = -2$$

